MATHIS VIGNEAU

MECHANICAL ENGINEERING



mathis.vigneau@gmail.com
+1 (914) 386-7380
LinkedIn profile
Portfolio:
https://github.com/mathis4612/portifolio_mathis

EDUCATION

Cornell University

Bachelor's in Mechanical Engineering (Expected May 2027)

- Fluid Mechanics, Thermodynamics, Heat transfer, System dynamics, Mechanics of Engineering Materials, DFMEA Design Failure Mode Analysis, Differential Equations, Linear algebra, Mechatronics, Physics 1, 2, & 3, Applied Probability and Stats,

EXPERIENCES

Mechanical Engineering Intern, Binnies Hong Kong -Summer 2025

- Assisted in designing pump system control diagrams and cost estimates for major WSD projects, cutting down 3 months of work in the schedule.
- Converted complex mechanical schematics into clear presentation figures and organized technical documentation.
- Supported MVAC, plumbing, and fire services design, producing accurate material and labor estimates. Saving 12% of the original cost estimation

Mechanical Engineering Intern, PCI Pharma -Summer 2026

- Developed and revised SOPs and preventive maintenance procedures for lyophilization, sterilization, and manufacturing equipment to improve compliance, reliability, and operational safety.
- Developed a Sartocheck 5 filter-testing manual and training course, including troubleshooting and failure interpretation, enabling operators to identify filter, setup, and procedural failures independently.
- Troubleshot and repaired lyophilization and manufacturing equipment, resolving vacuum instability, hydraulic and electrical issues, and improving machine reliability.
- Created preventive maintenance plans, completion forms, training materials, and spare-part documentation while supporting equipment maintenance, inventory management, and manufacturing operations.

Cornell research March 2026 - Present

- Cornell Space Structures, where we creat novel structures in space to dramatically increase the capabilities of satellites and infrastructures in orbit.

Cornell ChemE Car Project Team mechanical lead
Feb 2024 - Dec 2026

- Designed and prototyped mechanical components in CAD for reliability and manufacturability.
- Collaborated with chemical and electrical sub-teams leads to enhance vehicle safety and performance.
- Supported fabrication, assembly, and testing of a chemically powered autonomous vehicle.

Supervisor Sales and Tech Support, Cornell Store
Jan 2024 - Present

- Supervise sales team
- Advise clients on IT hardware, software, and engineering tools.
- Provided hands-on troubleshooting
- Improve customer experience through technical guidance.

PROFILE

Outgoing, adaptable, and solutions-oriented individual with strong leadership abilities. Thrives in diverse teams and excels in high-pressure, fast-paced environments, staying organized, calm, and focused on delivering results.

SKILLS

Design & CAD Modeling: SolidWorks, AutoCAD, Fusion 360
Programming & Analysis: MATLAB, Ansys Granta Edupack, Ansys Mechanical, Excel, Word, PowerPoint, Canva,
Mechanical Skills: Thermodynamics & energy systems, structural analysis, rapid prototyping

LANGUAGES

French - native language
English - fluent
Spanish - intermediate
Dutch - intermediate

ENGAGEMENT

Cru, Community Group Leader
Aug 2024 – May 2026

Cornell Shakespeare, Actor
Sep 2024 – Febuary 2026

Author of "Why Engineering Matters" Published in 2022